

Week 4 Day 4:

Model Tuning, Calibration & Final Pipeline Report

1. Objective

The objective of this task was to improve the performance and reliability of the machine learning pipeline developed during the previous stages of the project. This involved performing hyperparameter tuning on the best candidate models, diagnosing model bias and variance using learning and validation curves, calibrating predicted probabilities, selecting an appropriate classification threshold based on the project's business objective, and creating a fully reproducible pipeline suitable for deployment.

2. Hyperparameter Search

Hyperparameter optimization was performed using **RandomizedSearchCV** with **5-fold Stratified Cross-Validation** (`shuffle=True, random_state=42`). Precision was selected as the primary evaluation metric because the project's objective prioritized reducing false positive predictions.

Three candidate models were tuned:

- Logistic Regression
- HistGradientBoosting Classifier
- LightGBM Classifier

The following hyperparameters were explored:

Model	Hyperparameters Tuned
Logistic Regression	C, L1/L2 Regularization (<code>l1_ratio</code>)
HistGradientBoosting	learning_rate, max_depth, max_iter, l2_regularization
LightGBM	learning_rate, max_depth, n_estimators, reg_lambda

Each model was evaluated using **50 randomly sampled hyperparameter combinations**, providing a balance between search quality and computational cost.

Among the tuned models, **LightGBM achieved the highest cross-validated Precision** and was selected as the final model for subsequent analysis and evaluation.

3. Model Diagnostics

Model performance was evaluated using learning curves and validation curves to determine whether the selected models exhibited underfitting or overfitting.

Learning Curve

The learning curve for the tuned LightGBM model showed that both the training and validation precision increased as additional training samples were used. The gap between the two curves remained small throughout the experiment, indicating that the model generalized well to unseen data and showed no significant evidence of overfitting. Validation performance gradually stabilized with increasing training size, suggesting that the model had largely converged and that adding substantially more data would likely result in only modest improvements.

Logistic Regression Validation Curve

The validation curve for Logistic Regression examined the effect of the regularization parameter (C) on model performance. Very small values of C produced lower training and validation precision due to stronger regularization, indicating slight underfitting. As C increased, validation precision improved before stabilizing, while the difference between training and validation performance remained minimal. This suggests that the tuned Logistic

Regression model achieved an appropriate balance between model complexity and generalization without exhibiting significant overfitting.

LightGBM Validation Curve

The validation curve for LightGBM examined the effect of **max_depth**. Very shallow trees (low max_depth values) underfit the training data, resulting in lower training and validation precision. Increasing the tree depth substantially improved performance until approximately moderate depth values, after which both curves plateaued. The consistently small gap between training and validation precision indicates that the tuned LightGBM model generalized well and did not exhibit significant overfitting.

4. Probability Calibration and Threshold Selection

The probability estimates produced by the final LightGBM model were calibrated using **Isotonic Regression** through **CalibratedClassifierCV**. Calibration performance was evaluated using both the calibration curve and the Brier Score.

The calibrated model followed the ideal calibration line more closely than the uncalibrated model, indicating that the predicted probabilities more accurately reflected the true likelihood of the positive class. The improvement in the Brier Score further confirmed that calibration produced more reliable probability estimates.

Since **Precision** was selected as the primary evaluation metric throughout the project, the classification threshold was chosen to satisfy a business requirement of achieving **at least 75% Precision**. Instead of selecting the threshold that maximized the F1-score, the **lowest threshold achieving the target Precision was selected**. This approach preserves as much recall as possible while maintaining the desired level of Precision.

The selected operating threshold was:

Threshold = 0.45

5. Final Model Evaluation

The final calibrated LightGBM pipeline was evaluated on the previously untouched hold-out test set to estimate its expected performance on unseen data.

The complete preprocessing pipeline, feature engineering steps, probability calibration, and trained classifier were evaluated as a single integrated workflow.

Final Test Performance:

Metric	Value
Accuracy	0.8603
Precision	0.7456
Recall	0.6323
F1-score	0.6843
ROC-AUC	0.9129
PR-AUC	0.7763
Brier Score	0.0913

The final tuned and calibrated LightGBM pipeline achieved an **Accuracy of 86.03%** and a **Precision of 74.56%** on the untouched hold-out test set, meeting the project's primary evaluation objective of minimizing false positive predictions. The model also achieved a **Recall of 63.23%**, resulting in an **F1-score of 68.43%**. Strong discrimination performance was demonstrated by a **ROC-AUC of 0.9129** and a **PR-AUC of 0.7763**. After probability calibration, the selected operating threshold of **0.45** successfully satisfied the target precision

while preserving as much recall as possible, making the model suitable for deployment in a precision-focused classification scenario.

6. Final Pipeline and Reproducibility

The complete machine learning workflow was implemented using **scikit-learn Pipelines**, ensuring that feature engineering, preprocessing, calibration, and prediction were executed consistently during both training and inference.

To ensure reproducibility:

- Random seeds (`random_state=42`) were fixed throughout the project.
- Cross-validation splits were reproducible.
- Library versions were documented.
- The final trained pipeline was serialized using **Joblib**, allowing it to be loaded later without retraining.

This approach ensures that future predictions use the exact same preprocessing and model configuration as the training pipeline.

7. Conclusion

Hyperparameter tuning improved the performance of the candidate models while maintaining robust generalization across cross-validation folds. Diagnostic learning and validation curves showed minimal evidence of overfitting and confirmed that the selected LightGBM model achieved a good balance between model complexity and predictive performance.

Probability calibration further improved the reliability of the predicted probabilities, making them more suitable for threshold-based decision making. Because Precision was the primary business objective throughout this project, the final operating threshold was selected to achieve a minimum Precision of **75%**, resulting in a deployment threshold of **0.45**.

The final solution is a fully reproducible and deployable machine learning pipeline that integrates preprocessing, feature engineering, probability calibration, and classification into a single serialized artifact capable of generating consistent predictions on unseen data.